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| **PI Name:** | Grigoriev, Roman Ph.D. | | |
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| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Interfacial phenomena | | |
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| **Task Description:** | We propose studies of driven thin liquid films that exhibit wetting/spreading and capillary-driven phenomena cited by the National Research Council (NRC) as 'high-priority' research directions for NASA's Office of Biological and Physical Research (OBPR). The proposed study will address unsolved problems involving moving contact lines by combining theoretical modeling with experiments that can probe flow dynamics in ways that heretofore were not possible. This study uses a newly-developed experimental technique to manipulate optically (via the Marangoni effect) a broad class of capillary-dominated flows simultaneously at multiple spatial locations. With this technique, well-defined initial conditions can be imposed on spreading liquid films to probe dynamics and feedback can be applied to enable control. The experiments will examine spreading driven by Marangoni effects, which are ubiquitous in microgravity. Moreover, experiments will also examine gravitationally-driven spreading on inclined surfaces to gain insight into effects that may arise in Human Exploration and Development of Space (HEDS) technologies designed to operate in reduced gravity on other planets or in 'artificial' gravity produced by rotation. For all cases, optical actuation will be used to perturb the spreading films using disturbances with well-defined wave numbers, thereby permitting quantitative measurement of the growth/decay rates. Our ability to impose well-controlled disturbances in experiments will also be used to measure transient amplification effects, thereby enabling, for the first time in any fluid-dynamic system, quantitative theoretical/experimental examination of the controversial role that transients play in instability and transition. These results will be compared with predictions from different lubrication models using both linear stability analysis via eigenvalues and generalized stability analysis to characterize predicted transient growth due to the 'non-normal' character of linear operators that describe spreading films, allowing resolution of currently existing discrepancies between experimental measurements and theoretical predictions of growth rates and transient amplification factors. Additionally, both theory and experiment will be used to investigate the role of nonlinear terms in transition and the relation between transient amplification and secondary instability of a weakly perturbed flow, which offer alternative explanations for transition. Finally, optical actuation will be used to provide feedback in response to flow measurements, thereby permitting investigation of methods for selecting desired flow states, either by selective patterning of the flow or by suppression of instability. The results of the proposed work should contribute strongly to the fundamental, strategic and commercial research thrusts of OBPR. Our work emphasizes the use of optical actuation as a unique tool for studying fundamental scientific problems in liquid films that not only address the OBPR Physical Sciences Research Division's Fluid Physics interests in dynamics, instabilities and interfacial phenomena, but also connect, in novel ways, to broader issues in Fluid Physics, including the transition to turbulence. The results from the proposed work will also be strategically important for evaluating the potential for success or failure of proposed HEDS technologies. The dynamics of wetting and spreading (recently ranked by the NRC as 'most important' in strategic impact relative to other high priority areas across all microgravity disciplines) play a crucial role in liquid flow management in a vast array of HEDS technologies for power generation, propulsion, and life support. The spreading behavior of liquids is of central importance in applications as diverse as the storage of cryogenic propellants, the suppression of fire aboard spacecraft, the rewetting of hot surfaces in space-based multiphase heat transfer devices and the cleanup of liquid spills in low gravity. The results from the proposed work should also be commercially relevant to devising practical methods of controlling thin film flows; in particular, our results could lay the foundation for a new generation of devices for controlling interfacial flows on earth as well as in space. Wetting and spreading dynamics (ranked by the NRC as 'most important' in commercial impact relative to other high priority areas across all microgravity disciplines) can strongly affect the coating of solid surfaces, a process that is ubiquitous in industry (e.g., spin-coating of semiconductor wafer) and are directly relevant to emerging technologies such as microfluidics devices for biological and chemical applications. Thus, in summary, our proposed work closely follows the recommendations of expert external committees by building upon long standing OBPR/PSRD strengths in directions that contribute most effectively and directly to OBPR/PSRD's support of NASA's mission. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Liquid spreading/wetting phenomena play an important role in numerous commercial terrestrial applications. For instance, contact-line dynamics controls the coating of solid surfaces, a process that is ubiquitous in industry (e.g., spin-coating of semiconductor wafers). Improved understanding of stability of driven contact lines will help impove the speed and quaility of the coating process. Another important application is to “emerging technologies” such as microfluidics, where termocapillary and electrowetting effects have been proposed as a basic physical mechanism for manipulating minute amounts of fluids. Our research paves the way to a new generation of dynamically reprogrammable microfluidic devices where the flow is controlled using optical means. |
| **Task Progress:** | To improve our understanding of the stability of moving contact lines we have conducted a theoretical investigation of transient amplification of disturbances using the long-wave length approximation. Our research has demonstrated that the disagreement in theoretical predictions for the magnitude of transient amplification obtained using different models of gravity-driven films was due to their improper analysis in many studies. In particular, we determined that the strength of transient growth can be severely underestimated when a computational domain of insufficient size is used to numerically solve the evolution equations. In some cases erroneous predictions were traced to disturbances, such as surface roughness or chemical heterogeneity, ahead of the contact line having been ignored. We have determined that it is precisely these types of disturbances that are amplified most strongly when the fluid passes over them. We have also found that the transient amplification magnitude depends sensitively on the streamwise and transverse structure of the disturbance and that some distrubances, although uncommon in experiments, could be amplified by many orders of magnitude over large spatial scales.<br>We have performed a nonlinear stability analysis for a uniform state of a model system closely related to the lubrication equations describing the dynamics of driven contact lines. We have discovered that, in the presence of nonlinearity and non-normality, disturbances with magnitude below a certain threshold transiently grow and then decay to the uniform state. Disturbances larger than the threshold diverge away from the uniform state. The critical size of disturbances was found to be controlled by the unstable equilibrium solutions emerging near the linearly stable uniform state and analogous to, e.g., Tollmien-Schlichting waves that undergo secondary instability in boundary shear flows. This mechanism can provide an alternative route to destabilization of the transversely uniform thin film flow, explaining the development of the contact line modulation for parameters for which a linearly stable evolution is predicted.<br><br>We have conducted an experimental investigation of transient dynamics of thermally driven thin films. Initial disturbances were imposed via distributed optical heating applied behind the contact line, creating local elevations and depressions in the film thickness. Transient amplification of these disturbances as a function of time was determined experimentally by taking the ratio of the norms of initial and evolved disturbances and factoring out exponential growth. Experimental results were compared with the predictions of the computational model derived based on a |

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| | similar model constructed earlier for the gravity-driven flows. Quantitative comparisons between experimentally observed and numerically computed transient amplification were found to be in reasonably good agreement.<br><br>Transient amplification is expected to be easier to measure in gravitationally driven thin films as the evolution is predicted to be linearly stable for small inclination angles. We have therefore constructed an experimental apparatus for studying gravitationally driven films and conducted preliminary experiments to investigate the qualitative behavior of thin films for various angles of inclination (with respect to the horizontal). |
| **Bibliography Type:** | Description: (Last Updated: 01/03/2008) |
| **Articles in Peer-reviewed Journals** | Handel A, Grigoriev RO. "Transient dynamics and nonlinear stability of spatially extended systems." Phys Rev E Stat Nonlin Soft Matter Phys. 2006 Sep;74(3 Pt 2):036302. Epub 2006 Sep 14. PMID: 17025738 , Sep-2006 |
| **Articles in Peer-reviewed Journals** | Grigoriev RO. "Transient growth in driven contact lines." Physica D 2005 Sep 15;209(1-4):105-16. , Sep-2005 |